

Log Processing Methods and Challenges in Academic Management Information Systems

Modern ICT infrastructures within academic information systems generate vast volumes of heterogeneous log data originating from applications, containers, operating systems, networks, and security components. Although these logs provide fine-grained operational and security observations, they are typically produced in raw, unstructured, and source-specific formats. This lack of standardization and semantic alignment significantly complicates cross-system analysis, event correlation, and the reconstruction of operational or security incidents, particularly in complex and distributed environments.

To address these challenges, this study adopts a progressive abstraction model in which raw log entries are first interpreted as events, events are semantically linked into traces, and traces are further composed into higher-level narratives that describe meaningful operational states or security situations (Raw Logs → Events → Traces → Narratives). This layered abstraction mirrors the cognitive processes employed by practitioners when diagnosing system failures, conducting root-cause analysis, or investigating security incidents.

The primary objective of this survey is to elicit and validate competency questions that reflect real-world analytical needs at each level of abstraction. Specifically, the survey aims to:

- Validate the extent to which the proposed abstraction pipeline aligns with current operational and security practices in academic information systems;
- Identify semantic, structural, and contextual gaps that hinder effective event correlation, trace reconstruction, and situational understanding;
- Assess the perceived value of a semantic log ontology in supporting automated reasoning, cross-source correlation, and incident reconstruction.

Participants are asked to evaluate the relevance and importance of each competency question (CQ), enabling their prioritization during ontology design and refinement. The survey is intended to be validated by domain experts in system operations, security analysis, and incident response, and also provides participants with the opportunity to propose additional questions, abstraction levels, or dimensions that they consider relevant.

Thank you for taking part in this survey. Based on responses received so far, it should take approximately 10–15 minutes to complete.

*** Indica uma pergunta obrigatória**

1. Email *

Identification of the Institution or Organization

2. Type of Institution or Organization *

Marcar apenas uma oval.

☐ Public

☐ Private

☐ Other

3. Which sector best describes your primary institutional affiliation? *

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☐ Academia (universities or research institutions)

☐ Defense or national security organizations

☐ Government or public sector institutions

☐ Industry (private sector organizations)

☐ Other

4. What Academic Management Information System is currently used by your Institution or Organization? *

5. Contact point name (survey respondent)

6. Email for future contact

Questionnaire

A – Context and Log Ecosystem

7. Do you perform correlation of log data across multiple sources or systems as part of your current analysis workflow? *

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☐ Yes

☐ No

8. If **Yes** which of the following approaches do you use to correlate log data across different sources or systems? (**Select all that apply**) *

Marcar tudo o que for aplicável.

- ☐ Manual inspection (e.g., grep-based searches, timestamp alignment)
- ☐ Centralized logging platforms (e.g., ELK stack, Splunk, Graylog)
- ☐ Distributed tracing or APM tools (e.g., Jaeger, Zipkin, OpenTelemetry)
- ☐ SIEM, SOAR, or security analytics platforms
- ☐ Custom scripts or ad hoc tooling
- ☐ I do not perform cross-source correlation
- ☐ Other

9. If **No** what is the primary reason you do not perform cross-source log correlation? (**Select all that apply**) *

Marcar tudo o que for aplicável.

- ☐ Lack of tooling or infrastructure
- ☐ Insufficient data integration
- ☐ Limited perceived benefit
- ☐ Resource or time constraints
- ☐ Not applicable to my role
- ☐ Other

10. Do you rely on specific log sources as critical inputs for your routine analytical activities? *

Marcar apenas uma oval.

- ☐ Yes
- ☐ No

11. Which of the following log sources do you consider critical for your analytical tasks? (**Select all that apply**) *

Marcar tudo o que for aplicável.

- ☐ Application logs
- ☐ Security and audit logs
- ☐ System and kernel logs
- ☐ Network logs
- ☐ Cloud infrastructure logs
- ☐ The relevance depends on the incident type
- ☐ Other

12. What limitations or challenges do you encounter when reconstructing events from log data? (**Select all that apply**) *

Marcar tudo o que for aplicável.

- ☐ Absence of common identifiers across log sources
- ☐ Inconsistent or heterogeneous log formats
- ☐ Missing, incomplete, or corrupted log data
- ☐ Time synchronization and timestamp alignment issues
- ☐ High log volume or excessive noise
- ☐ Significant manual effort required for correlation
- ☐ Limitations of existing tools or platforms
- ☐ Other

B – Event Discovery & Filtering

13. When searching for events within a specific time range, which filters do you typically apply? (**Select all that apply**) *

Marcar tudo o que for aplicável.

- ☐ Time window (start/end time)
- ☐ Host / node / machine
- ☐ Application or service name
- ☐ Severity / log level (e.g., INFO, WARN, ERROR)
- ☐ User or account identity
- ☐ Request / correlation ID
- ☐ IP address (source or destination)
- ☐ Event type or category
- ☐ Component / subsystem (e.g., auth, DB, network)
- ☐ Other

14. How do you identify which events belong to a specific application, service, or host? *

Marcar apenas uma oval.

- ☐ Explicit application/service field
- ☐ Hostname or node identifier
- ☐ Container or pod identifier
- ☐ Log source or file path
- ☐ Manual inference from message content
- ☐ Often difficult or ambiguous

15. How important is it to group events belonging to the same request or execution flow? *

Marcar apenas uma oval.

	1	2	3	4	5	
Not	☐	☐	☐	☐	☐	Critical

16. Which identifiers do you use to group events belonging to the same request or execution flow? (**Select all that apply**) *

Marcar tudo o que for aplicável.

- ☐ Request ID
- ☐ Correlation ID
- ☐ Session ID
- ☐ Trace ID (distributed tracing)
- ☐ User ID
- ☐ IP address
- ☐ Timestamp proximity (time-based heuristics)
- ☐ None / rarely available
- ☐ Other

C – Event Lineage & Tracing

17. How do you currently reconstruct the sequence leading to an error or incident? *

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- ☐ Manual timeline inspection of logs
- ☐ Correlation using request identifiers or trace IDs
- ☐ Use of distributed tracing or application performance monitoring (APM) tools
- ☐ Use of SIEM-based correlation rules
- ☐ Predominantly ad hoc or case-by-case approaches
- ☐ I am usually unable to reconstruct complete event sequences

18. How useful is access to log data representing events occurring *before* and *after* an error or incident for your analysis? *

Marcar apenas uma oval.

	1	2	3	4	5	
Not	☐	☐	☐	☐	☐	Extremely useful

19. How do you typically correlate security events with system or application events? (**Select all that apply**) *

Marcar tudo o que for aplicável.

- ☐ Shared identifiers (e.g., user ID, IP address, session ID)
- ☐ Time-based correlation (temporal proximity)
- ☐ SIEM correlation rules or detection logic
- ☐ Manual reasoning or ad-hoc investigation
- ☐ Automated tracing or correlation tools (non-SIEM)
- ☐ I rarely or never correlate these events
- ☐ Other

20. How do you correlate security events with system or application events in your current analysis workflow? *

Marcar apenas uma oval.

- ☐ Correlation using shared identifiers (e.g., user IDs, IP addresses, session or request identifiers)
- ☐ Time-based correlation (e.g., timestamp alignment or temporal proximity)
- ☐ Use of SIEM-based correlation rules or analytics
- ☐ Predominantly manual reasoning and contextual analysis
- ☐ Such correlation is rarely or not feasible in practice

D – Authentication & Access Tracing

21. How do you analyze authentication attempts preceding access or privilege escalation events? *

Marcar apenas uma oval.

- ☐ Manual review of authentication logs
- ☐ Correlation of authentication logs with audit or access-control logs
- ☐ Use of SIEM rules, alerts, or automated analytics
- ☐ Analysis is performed infrequently or only in response to alerts
- ☐ I do not typically perform this type of analysis

22. Which types of information are required to accurately reconstruct a complete user session timeline? **(Select all that apply)** *

Marcar tudo o que for aplicável.

- ☐ User identifier (e.g., username or account ID)
- ☐ Session identifier (e.g., session ID or token)
- ☐ Source network information (e.g., IP address)
- ☐ Authentication method or mechanism used
- ☐ Actions performed or commands executed
- ☐ Timestamps and temporal ordering information
- ☐ Target host, system, or application accessed

23. How do you determine whether multiple SSH-related events belong to the same session? *

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- ☐ Use of explicit session identifiers
- ☐ Correlation based on source IP address and user account
- ☐ Use of terminal or TTY identifiers
- ☐ Time-based heuristics (e.g., inactivity thresholds)
- ☐ Manual inspection of log entries
- ☐ It is often unclear or indeterminable

24. Which identifiers do you consistently rely on during log analysis and event correlation? **(Select all that apply)** *

Marcar tudo o que for aplicável.

- ☐ User identifier (e.g., user ID or account name)
- ☐ Session identifier
- ☐ Network identifier (e.g., IP address)
- ☐ Terminal or TTY identifier
- ☐ Process identifier (PID)
- ☐ Host or system identifier (e.g., hostname)
- ☐ None of the above are used consistently

E – Application & Container Tracing

25. How do you typically detect links between container lifecycle events and application errors? **(Select all that apply)** *

Marcar tudo o que for aplicável.

- ☐ Shared container or pod identifiers (e.g., container ID, pod name)
- ☐ Time-based correlation (temporal proximity)
- ☐ Monitoring or alerting tools (e.g., Kubernetes, APM, observability platforms)
- ☐ Automated correlation or tracing mechanisms
- ☐ Manual inspection or ad-hoc analysis
- ☐ It is usually difficult to establish such links
- ☐ Other

26. What information helps you correlate database events with application requests? (**Select all that apply**) *

Marcar tudo o que for aplicável.

- ☐ Request or transaction ID
- ☐ Application or service name
- ☐ User or account identity
- ☐ Query identifiers (e.g., query ID, statement hash)
- ☐ Connection or session identifiers
- ☐ Time-based correlation (timestamp proximity)
- ☐ It is rarely or not possible to correlate these events
- ☐ Other

F – System-Level Tracing

27. Which system-level events do you commonly observe before system instability? (**Select all that apply**) *

Marcar tudo o que for aplicável.

- ☐ Kernel warnings or errors
- ☐ Resource exhaustion (CPU, memory, disk)
- ☐ Network errors or connectivity issues
- ☐ Time synchronization events (e.g., NTP adjustments)
- ☐ Hardware-related events (e.g., disk, memory, sensors)
- ☐ Operating system or service restarts
- ☐ I am not sure / I have not observed clear precursors

28. How important is visibility into package installation and update events for your analysis? *

Marcar apenas uma oval.

1 2 3 4 5

Not ☐ ☐ ☐ ☐ ☐ Extremely important

29. When package updates cause issues, what evidence do you typically look for? *
(Select all that apply)

Marcar tudo o que for aplicável.

- ☐ Package installation or update logs (e.g., apt, dpkg, yum)
- ☐ Version changes or downgrade/upgrade history
- ☐ Service restart or reload events
- ☐ Error or warning logs occurring after the update
- ☐ Dependency or compatibility conflicts
- ☐ Configuration changes triggered by the update
- ☐ It is usually hard to determine the root cause
- ☐ Other

G – Security Tracing

30. For sensitive operations (e.g., file access, privilege escalation, configuration changes), what audit information do you require? *
(Select all that apply)

Marcar tudo o que for aplicável.

- ☐ User or service identity
- ☐ Command or action performed
- ☐ Target resource (file, service, configuration, secret)
- ☐ Timestamp
- ☐ Outcome (success, failure, partial)
- ☐ Source IP address or origin
- ☐ Authentication method or context
- ☐ None / minimal audit information required
- ☐ Other

31. How do you typically correlate secret-management events with authentication or authorization events? **(Select all that apply)** *

Marcar tudo o que for aplicável.

- ☐ Shared user or service identity
- ☐ Time-based correlation (temporal proximity)
- ☐ SIEM or security platform correlation rules
- ☐ Automated correlation or tracing tools (non-SIEM)
- ☐ Manual analysis or investigation
- ☐ It is rarely or not possible to correlate these events
- ☐ Other

32. What factors make security trace reconstruction difficult in your environment? **(Select all that apply)** *

Marcar tudo o que for aplicável.

- ☐ Logs distributed across multiple systems or platforms
- ☐ Missing or inconsistent identifiers across logs
- ☐ Encryption, masking, or redaction of relevant fields
- ☐ High log volume and noise
- ☐ Tooling or platform limitations
- ☐ Limited retention or incomplete log coverage
- ☐ Lack of standardized log semantics
- ☐ Other

H – Cross-Source Log Correlation

33. How do you correlate Syslog/Journalctl with application logs? *

Marcar apenas uma oval.

- ☐ Time-based correlation (timestamp proximity)
- ☐ Host or service identifiers (hostname, service name)
- ☐ Centralized log or observability platform
- ☐ Shared request, session, or trace identifiers
- ☐ Automated correlation or tracing tools
- ☐ Manual inspection or ad-hoc analysis
- ☐ It is rarely or not possible to correlate these logs
- ☐ Other

34. Which recurring multi-source event patterns do you look for when diagnosing issues or incidents? (**Select all that apply**) *

Marcar tudo o que for aplicável.

- ☐ Authentication events → errors → alerts
- ☐ Deployment or release events → errors → rollback
- ☐ Resource exhaustion → service or system failures
- ☐ Configuration changes → anomalies or unexpected behavior
- ☐ Privilege escalation → system or configuration changes
- ☐ Lateral movement or unusual access → follow-up system activity
- ☐ I do not typically look for recurring multi-source patterns
- ☐ Other

35. What factors make cross-system or cross-service log correlation difficult in your environment today? (**Select all that apply**) *

Marcar tudo o que for aplicável.

- ☐ Lack of shared identifiers across systems
- ☐ Inconsistent or ambiguous log semantics
- ☐ Tool fragmentation across teams or platforms
- ☐ High manual effort required for correlation
- ☐ High data volume or noise
- ☐ Limited cross-system visibility or access
- ☐ Performance or scalability limitations of tools
- ☐ Other

I – Anomaly & Incident Reconstruction

36. What information is essential for reconstructing an anomaly or incident? (**Select all that apply**) *

Marcar tudo o que for aplicável.

- ☐ Timeline of relevant events
- ☐ Affected users, services, or assets
- ☐ Related system or configuration changes
- ☐ Security context (authentication, authorization, access attempts)
- ☐ Root-cause indicators or hypotheses
- ☐ Impact or scope assessment
- ☐ I am not sure / depends on the case
- ☐ Other

37. What are the biggest blockers you face when reconstructing incidents from logs? (**Select all that apply**) *

Marcar tudo o que for aplicável.

- ☐ Lack of effective correlation across logs
- ☐ Missing, incomplete, or inconsistent data
- ☐ Tooling or platform limitations
- ☐ Time constraints during investigation
- ☐ Skill or domain-knowledge gaps
- ☐ High data volume or noise
- ☐ Organizational or access constraints
- ☐ Other

J – Desired Features for a Semantic Log Ontology

38. Which concepts should always be standardized in a semantic log ontology? (**Select all that apply**) *

Marcar tudo o que for aplicável.

- ☐ Event type or category
- ☐ Actor (user, service, process)
- ☐ Target or affected resource
- ☐ Time and temporal context
- ☐ System, component, or service context
- ☐ Correlation identifiers (request, session, trace IDs)
- ☐ Outcome or result (success, failure, error)
- ☐ Severity or risk level
- ☐ None / minimal standardization required
- ☐ Other

39. Which types of automated reasoning would be most valuable in a semantic log ontology? **(Select all that apply)** *

Marcar tudo o que for aplicável.

- ☐ Automated event correlation across sources
- ☐ End-to-end trace reconstruction
- ☐ Root-cause analysis and causal inference
- ☐ Anomaly or deviation detection
- ☐ Security incident or attack-chain reconstruction
- ☐ Impact analysis (affected services, users, assets)
- ☐ None / limited automated reasoning needed
- ☐ Other

40. How would a unified semantic log model improve your workflow? **(Select all that apply)** *

Marcar tudo o que for aplicável.

- ☐ Faster investigations and triage
- ☐ Improved accuracy and context awareness
- ☐ Reduced manual effort
- ☐ Increased automation and scalability
- ☐ Better cross-team communication
- ☐ Minimal or no noticeable impact
- ☐ Other

41. Which compliance, audit, or security frameworks must your logging or monitoring model support? **(Select all that apply)** *

Marcar tudo o que for aplicável.

- ☐ SOC 2
- ☐ ISO 27001
- ☐ NIST (e.g., CSF, SP 800-series)
- ☐ PCI DSS
- ☐ GDPR or data protection regulations
- ☐ Internal security or audit policies
- ☐ Industry-specific frameworks (e.g., HIPAA, SOX)
- ☐ None / no specific framework requirements
- ☐ Other

K – Final Reflections

42. What are your top frustrations with log analysis today? **(Select all that apply)** *

Marcar tudo o que for aplicável.

- ☐ Excessive log volume or noise
- ☐ Poor or unreliable correlation across logs
- ☐ High manual effort required
- ☐ Missing context or insufficient metadata
- ☐ Tool fragmentation across teams or platforms
- ☐ Limited automation or intelligence
- ☐ Steep learning curve or complexity
- ☐ Other

43. What are the most valuable improvements a semantic log ontology could bring to your environment? **(Select all that apply)** *

Marcar tudo o que for aplicável.

- ☐ Unified, cross-source view of logs and events
- ☐ Automated correlation across systems and services
- ☐ Improved traceability from low-level events to high-level incidents
- ☐ Faster and more reliable root-cause analysis
- ☐ Improved security visibility and incident context
- ☐ Better support for compliance and auditing
- ☐ Reduced investigation time and effort
- ☐ None / limited expected improvement
- ☐ Other

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